

Additive forcing of a random-frequency oscillator: a catalogue of simulations, measurements and figures

Abstract

This document is an inventory. It lists every simulation run on the additively forced random-frequency oscillator, the parameter grid each one covered, the quantities measured, and every figure produced, together with an explicit guide to what is plotted on each axis and what each line, colour and marker represents. Numbers quoted in the text are the measured values. *No interpretation, verdict or conclusion is offered anywhere in this document*; the figures and the numbers are presented as they came out. Parametric forcing is excluded except where a parametric run was used purely to calibrate a detector, which is stated explicitly at that point.

Contents

1	The model, the observables and the numerical method	1
1.1	Equations	1
1.2	Integrator	2
1.3	The observables	2
2	Validation of the integrator	3
3	The blanket sweep of the reduced system	3
3.1	Where the response sits in frequency	4
3.2	Frequency-tracking tests	7
3.3	Trajectories and distributions	8
4	The full three-dimensional system	10
5	Calibration of the detector	10
6	Cycle length under the phase-winding definition	11
7	Cycle length under the zero-crossing definitions	15
7.1	The two counting rules	15
7.2	The fine sweep	16
7.3	Occupancy of the integer bands	18
7.4	Shape of the distribution at a fixed frequency	19
7.5	Sensitivity of the branch to the drive	20
8	Index of simulations	22
9	Index of figures	22
10	Index of measured values	23

1 The model, the observables and the numerical method

1.1 Equations

Everything below is run on one of two systems.

The reduced two-dimensional system. In polar-like coordinates (h, φ) , where h is an energy-like amplitude variable and φ a phase, the additively forced oscillator is integrated as

$$\ddot{X} + \varepsilon \dot{X} + X = \delta \sin(\omega t) + (\text{random-frequency term}), \quad (1)$$

with the random-frequency term reduced, in the white-noise limit, to a state-dependent diffusion of strength governed by σ and by the exponent parameter r . The stationary density of h in the *unforced* case is known in closed form,

$$p_s(h) = C(\sigma^2 + rh)^{-2/r}, \quad C = (2 - r)(\sigma^2)^{2/r-1}, \quad (2)$$

normalisable for $r < 2$ and of finite mean for $r < 1$.

The full three-dimensional system. Before the reduction, the frequency itself is an Ornstein–Uhlenbeck process A_t with correlation time τ , and the stiffness in (1) is A_t^2 . The reduced system is the $\tau \rightarrow 0$ limit. This system is used in §4.

Parameters.

ε	damping
σ	noise amplitude of the random-frequency term
r	multiplicative-noise exponent; $r = 0$ makes the system exactly linear
δ	forcing amplitude
ω	forcing frequency
τ	correlation time of A_t (three-dimensional system only)

The natural frequency is $\omega_0 = 1$ throughout, so the free period is $2\pi \approx 6.283$ and the drive period is $T_{\text{drive}} = 2\pi/\omega$.

1.2 Integrator

A symplectic (semi-implicit) Euler–Maruyama scheme is used, implemented in C and called through `ctypes`; the random number generator is `xoshiro256++`. Forward Euler–Maruyama was rejected because it inflates the energy by a factor $(1 + \Delta t^2)$ per step; the Milstein correction vanishes for this system, so the scheme is of strong order 1. When h exceeds 10^{12} the state is rescaled by 10^{-6} and the discarded factor is accumulated in $\log h$, so growth rates remain measurable without overflow. Common random numbers are used across a sweep: at fixed seed, every (δ, ω) cell sees the same noise realisation, so differences between cells are not seed noise.

Standard settings unless a section says otherwise: $\Delta t = 0.0025$, output thinned by 80 (sample interval 0.2), and a burn-in that is discarded before any statistic is computed.

1.3 The observables

Spectral peak of X . The largest local maximum of the periodogram of X . A “sub-drive” variant excludes frequency bands within $\pm 3\%$ of the first four drive harmonics $m\omega$, $m = 1, \dots, 4$, so that the free oscillation can be tracked even when the drive line dominates.

Tracking slope. $d(\text{response frequency})/d\omega$, fitted by least squares over a fixed window of ω ($[1.8, 2.2]$ for the 2:1 window, $[2.8, 3.2]$ for the 3:1 window). A response that follows the drive at half its frequency gives slope $\frac{1}{2}$; one that ignores the drive gives slope 0.

Phase coherence. $R_n = |\langle \exp i(\varphi - \omega t/n) \rangle|$ for $n = 1, 2, 3$, averaged over the run. Because the reference term depends on ω , the chance level does too, and nulls are matched on $|\omega/n - 1|$ rather than pooled.

Cycle period, phase-winding definition. The times at which the unwrapped phase φ advances by 2π ; successive differences are the cycle periods.

Cycle period, zero-crossing definitions. The times at which X crosses zero upward, found by linear interpolation between samples. Two counting rules are used and are compared in §7: *raw*, which accepts every upward crossing, and *hysteresis* (a Schmitt trigger), which accepts an upward crossing only if X has been below $-h_{\text{th}}$ since the previous accepted one, with $h_{\text{th}} = 0.25 \text{rms}(X)$.

N , cycle length in drive periods. $N = T_{\text{cycle}}/T_{\text{drive}} = T_{\text{cycle}} \omega/2\pi$. In the unforced system $T_{\text{cycle}} = 2\pi$ exactly, so $N = \omega$.

Growth rate. The least-squares slope of $\log h(t)$ against t , halved.

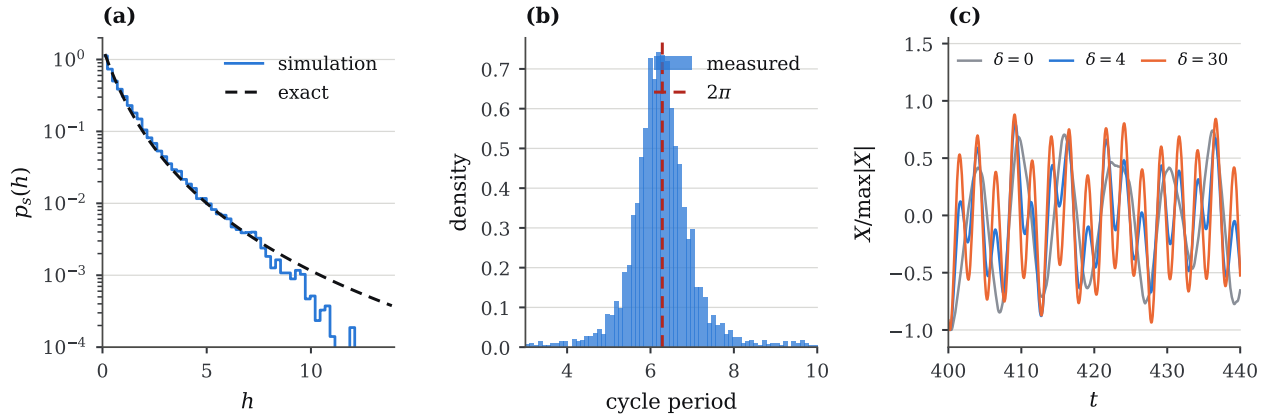
Sarle bimodality coefficient. $b = (g_1^2 + 1)/g_2$ from the skewness g_1 and kurtosis g_2 of the period distribution; $b = 5/9$ for a uniform distribution.

Band powers of $h(t)$. Fraction of the power of $h(t)$ falling in narrow bands centred on $\omega/3$, $\omega/2$, ω and 2ω .

2 Validation of the integrator

Simulation V1. Unforced reduced system, $\varepsilon = 0.05$, $\sigma = 1$, $r = 0.5$, $T = 20000$, burn-in 2000, single seed. Also, three short forced runs at $\omega = 2.5$ with $\delta \in \{0, 4, 30\}$, $T = 600$.

Measured. The simulated stationary density of h against the closed form (2); the distribution of the phase-winding cycle period against the free value 2π ; and three sample trajectories. Separately, the portable NumPy version of the integrator was checked against `scipy.integrate.solve_ivp` on the deterministic skeleton: growth rates agree to $\sim 10^{-5}$.



F01 — integrator validation. (a) Stationary density of the energy variable h . Horizontal axis h , vertical axis $p_s(h)$ on a logarithmic scale. Blue step curve: histogram from the simulation. Black dashed curve: the closed form (2). (b) Distribution of the cycle period measured from phase winding in the unforced system. Horizontal axis: cycle period; vertical axis: probability density. Red dashed vertical line: 2π , the free period. (c) Three trajectories over $t \in [400, 440]$ at $\omega = 2.5$, each divided by its own peak $|X|$ so that the shapes can be compared: grey $\delta = 0$, blue $\delta = 4$, orange $\delta = 30$.

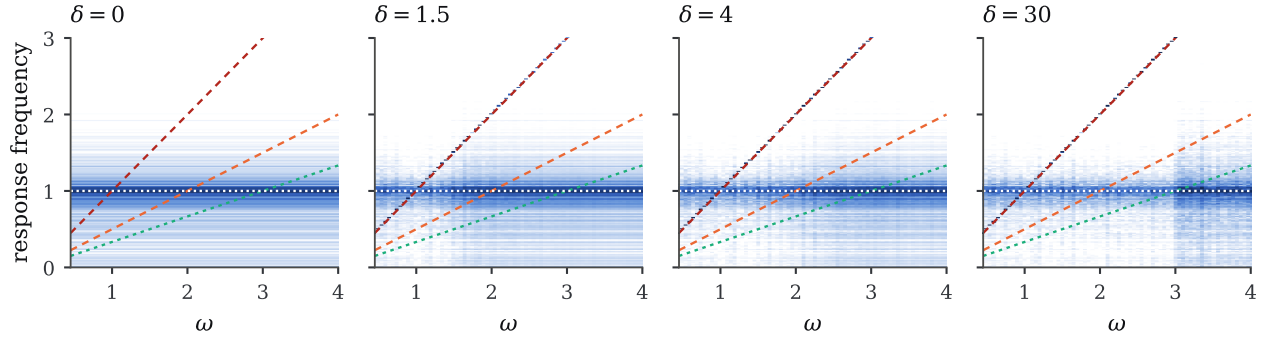
3 The blanket sweep of the reduced system

Simulation S1 — the main additive sweep. The grid is

ω	0.500, 0.525, ..., 4.000 (141 values, step 0.025)
δ	0, 0.5, 1.5, 4, 12, 30 (6 values; $\delta = 0$ is the in-grid null)
r	0, 0.25, 0.5, 0.9, 1.4 (5 values; $r = 0$ is exactly linear)
σ	0.5, 1, 2
ε	0.02, 0.05, 0.1, 0.2
seed	11, 22, 33, 44

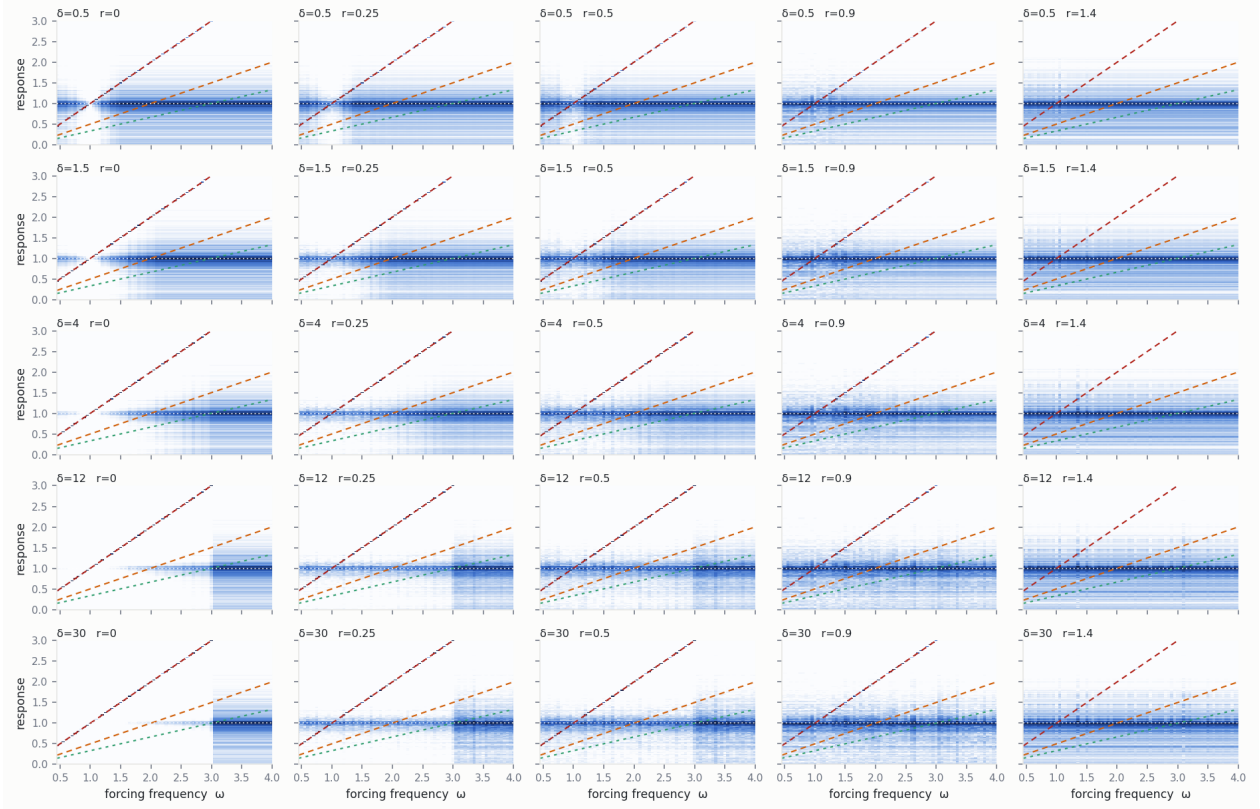
with $T = 3000$, $\Delta t = 0.0025$, burn-in 1000, sample interval 0.2. This is **203,040** runs in total, of which **33,840** have $\delta = 0$ and form the null. Grouping by $(\varepsilon, \sigma, r, \delta, \text{seed})$ gives **1,440** parameter cells, each a full curve in ω ; 1,200 of these have $\delta > 0$. Every run stores the three largest global spectral peaks, the two largest sub-drive peaks, R_1, R_2, R_3 , the four band powers of h , the number of cycles, the mean, standard deviation, skewness, kurtosis and bimodality coefficient of the period distribution, the 19 vigintile quantiles of that distribution, and h statistics with the growth rate. Output: `sweep_all.csv`. No verdict is stored in the file.

3.1 Where the response sits in frequency



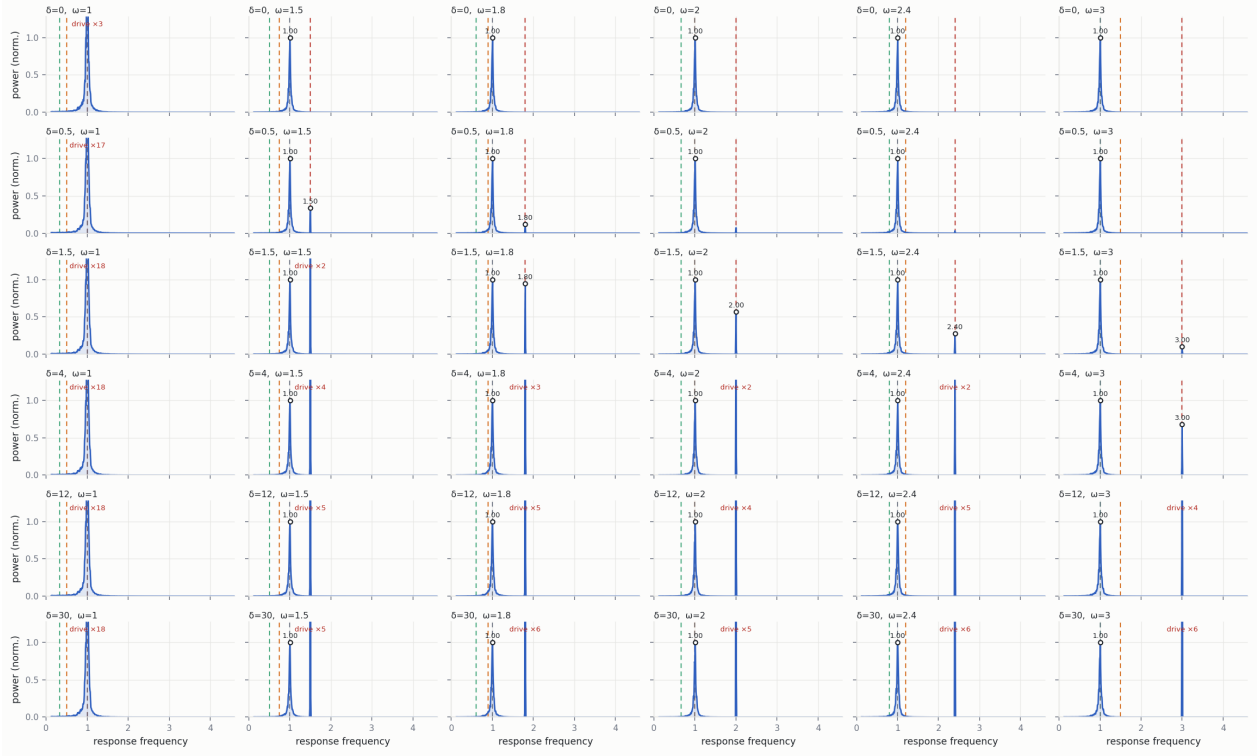
F02 — response spectrum against forcing frequency, four forcing strengths. Each panel is a two-dimensional map. Horizontal axis: forcing frequency ω , swept 0.45 to 4.00 in steps of 0.05. Vertical axis: response frequency, 0 to 3. Colour: $\log_{10}$ of the normalised spectral power of X at that response frequency, on a common scale from $10^{-3.4}$ (pale) to 1 (dark). Each vertical stripe of colour is therefore one run's spectrum, stood on end. The overlaid guides mark where features would fall: red dashed = ω (the drive itself), orange dashed = $\omega/2$, green dotted = $\omega/3$, white dotted = $\omega_0 = 1$ (the free frequency). Panels left to right: $\delta = 0, 1.5, 4, 30$. Settings $\varepsilon = 0.05$, $\sigma = 1$, $r = 0.5$, $T = 4000$, burn-in 1500.

F03 — Additive forcing: response spectrum of X against forcing frequency ω , over a 5×5 grid of (δ, r) . $\epsilon=0.05$, $\sigma=1$. Colour is $\log_{10}$ of normalised spectral power. Guides: red ω , orange $\omega/2$, green $\omega/3$, white $\omega_0=1$.



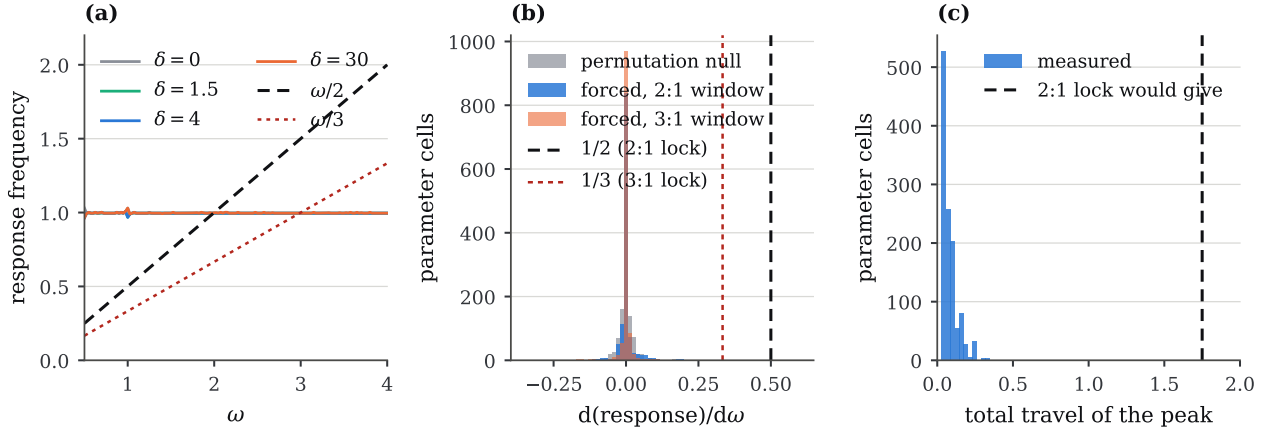
F03 — the same map over a 5×5 grid of (δ, r) . Rows are forcing strength $\delta \in \{0.5, 1.5, 4, 12, 30\}$ from top to bottom; columns are the noise exponent $r \in \{0, 0.25, 0.5, 0.9, 1.4\}$ from left to right. Each panel is read exactly as a panel of F02: horizontal axis ω from 0.45 to 4.00, vertical axis response frequency from 0 to 3, colour $\log_{10}$ power on a common scale. The guides are the same: red ω , orange $\omega/2$, green $\omega/3$, white $\omega_0 = 1$. The leftmost column, $r = 0$, is the exactly linear control. Settings $\epsilon = 0.05$, $\sigma = 1$, $T = 4000$, burn-in 1500.

F06 — Welch power spectra of X under additive forcing, $\varepsilon=0.05$, $\sigma=1$, $r=0.5$. Rows are forcing strength δ , columns are forcing frequency ω . Power is normalised to the largest peak OUTSIDE the drive band; where the drive peak exceeds the frame it is annotated by its factor. Vertical lines: red ω , orange $\omega/2$, green $\omega/3$, grey $\omega_0=1$.



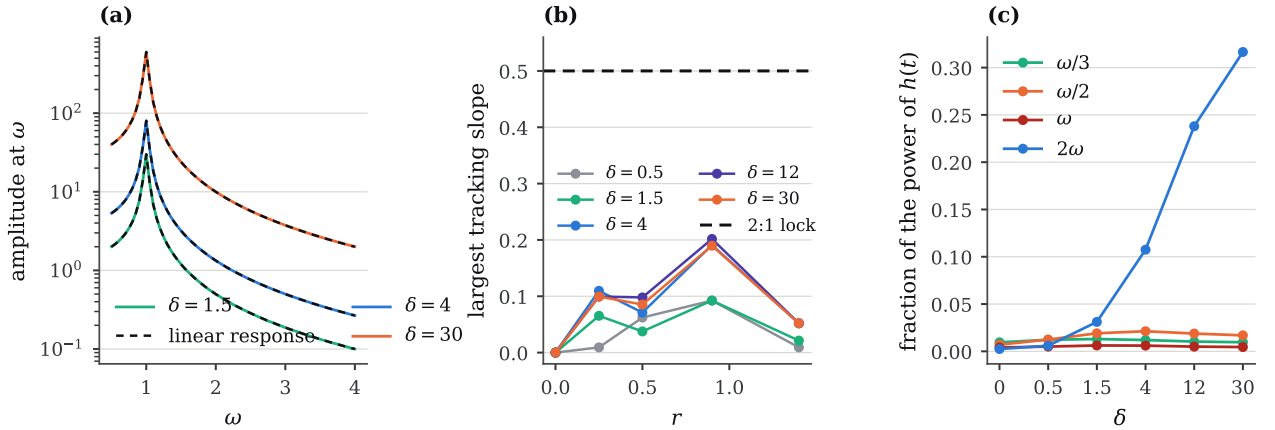
F06 — individual Welch spectra. A 6×6 grid of single runs. Rows: $\delta \in \{0, 0.5, 1.5, 4, 12, 30\}$. Columns: $\omega \in \{1, 1.5, 1.8, 2, 2.4, 3\}$. In every panel the horizontal axis is response frequency, 0 to 4.6, and the vertical axis is Welch power, *normalised to the largest peak lying outside the drive band* ($0.97\omega, 1.03\omega$) so that features other than the drive remain visible on a linear scale. Blue curve with shading: the spectrum. Open circles with labels: automatically detected peaks, annotated with their frequency to two decimals; peaks taller than 1.25 on the normalised scale are left unlabelled and instead the drive is annotated with the factor by which it exceeds the frame (“drive $\times n$ ”). Vertical dashed lines: red ω , orange $\omega/2$, green $\omega/3$, grey $\omega_0 = 1$. Settings $\varepsilon = 0.05$, $\sigma = 1$, $r = 0.5$, $T = 24000$, burn-in 4000.

3.2 Frequency-tracking tests



F04 — tracking-slope statistics over the whole sweep. (a) Median response frequency against ω , one curve per forcing strength: grey $\delta = 0$, green $\delta = 1.5$, blue $\delta = 4$, orange $\delta = 30$. The two reference lines are the loci a response would follow if it moved with the drive: black dashed $\omega/2$, red dotted $\omega/3$. (b) Histogram of the tracking slope, one count per parameter cell. Blue: the 1,200 forced cells fitted over $\omega \in [1.8, 2.2]$. Orange: the same cells fitted over $\omega \in [2.8, 3.2]$. Grey: the permutation null, obtained by randomly permuting the response values within each cell before fitting, which destroys any relation to ω while preserving the marginal spread. Vertical lines mark the values a 2:1 response ($\frac{1}{2}$, black dashed) and a 3:1 response ($\frac{1}{3}$, red dotted) would produce. (c) Histogram of the total travel of the peak, $\max - \min$ of the response frequency across the full ω range within a cell. The black dashed line at 1.75 is the travel a response following $\omega/2$ across this range would show.

Measured: mean slope in the 2:1 window $+0.0023$, maximum $+0.2017$; mean slope in the 3:1 window -0.0005 , maximum $+0.1084$; 0 of 1,200 cells reach half the 2:1 or 3:1 value. Median total travel 0.069, maximum 0.324.

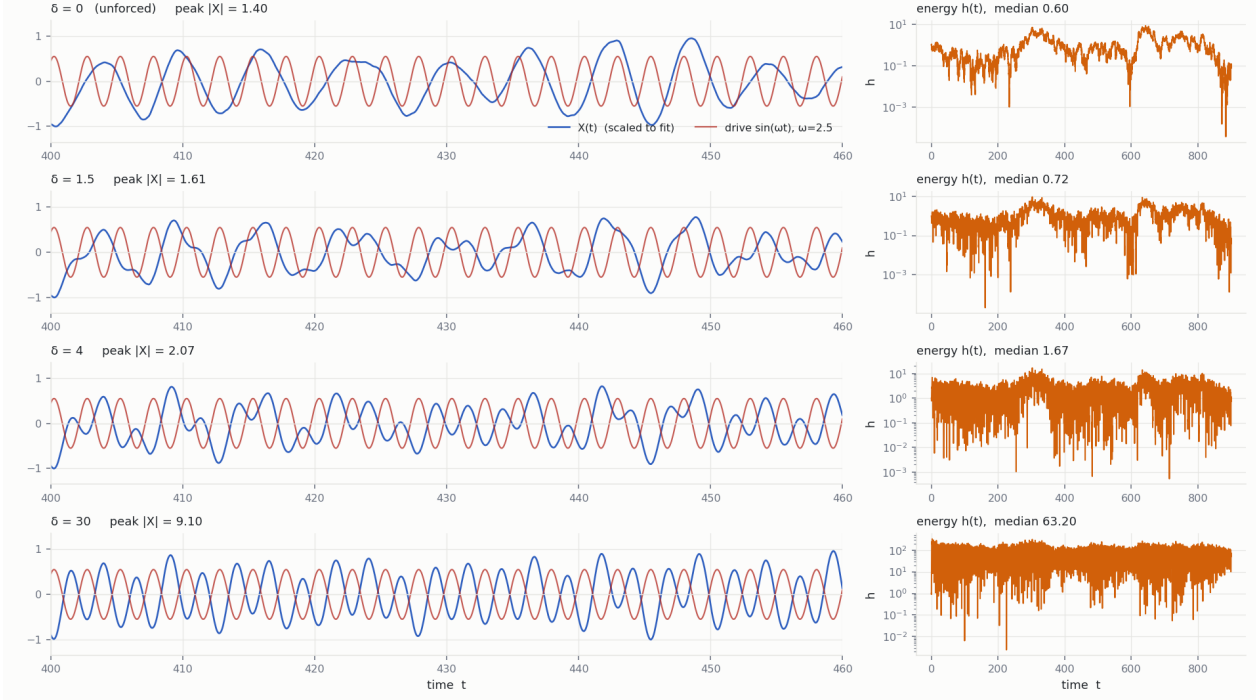


F05 — amplitude response, dependence on r , and the spectrum of h . (a) Amplitude of the component of X at exactly the drive frequency, extracted by projecting onto $\sin \omega t$ and $\cos \omega t$. Horizontal axis ω , vertical axis amplitude on a logarithmic scale. Solid curves: measured, green $\delta = 1.5$, blue $\delta = 4$, orange $\delta = 30$. Black dashed: the linear-response formula $A(\omega) = \delta / \sqrt{(1 - \omega^2)^2 + \varepsilon^2 \omega^2}$. These runs set $r = 0$. (b) The largest tracking slope found anywhere in the sweep, as a function of r . Horizontal axis r ; vertical axis the maximum over cells of the slope fitted on $\omega \in [1.8, 2.2]$. One curve per forcing strength. Black dashed line at $\frac{1}{2}$: the 2:1 value. (c) Fraction of the total power of $h(t)$ in narrow bands, against δ (categorical axis). Green $\omega/3$, orange $\omega/2$, red ω , blue 2ω .

Measured maximum tracking slope by r : 4.9×10^{-16} at $r = 0$; 0.1096 at $r = 0.25$; 0.0979 at $r = 0.5$; 0.2017 at $r = 0.9$; 0.0520 at $r = 1.4$. Power of h at 2ω : 0.25% at $\delta = 0$, rising to 31.7% at $\delta = 30$.

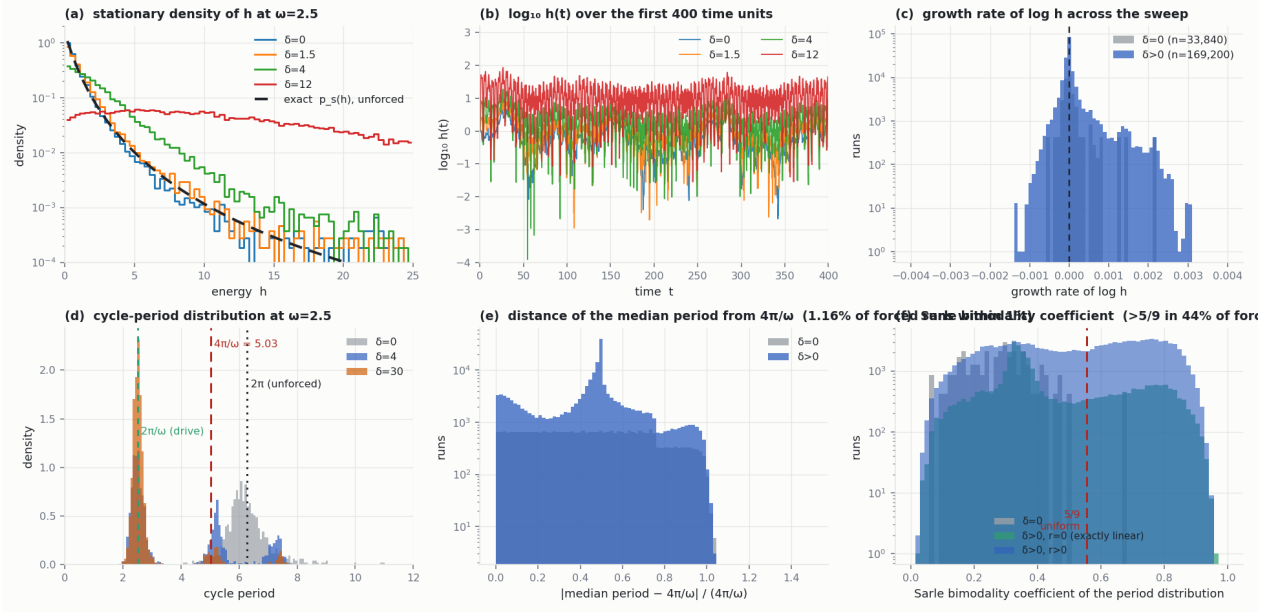
3.3 Trajectories and distributions

F07 — Sample paths at $\omega = 2.5$ for four forcing strengths. Left column: $X(t)$ over $t \in [400, 460]$, each panel rescaled to unit peak (blue), with the drive $\sin(\omega t)$ at fixed amplitude 0.55 (red). Right column: the energy $h(t)$ over the whole run, on a logarithmic axis. Reference periods: $2\pi/\omega = 1.257$ (drive), $4\pi/\omega = 2.513$, $2\pi \approx 6.283$ (unforced oscillator).



F07 — sample paths at $\omega = 2.5$. Four rows, one per forcing strength: $\delta = 0$ (unforced), 1.5, 4, 30. **Left column:** $X(t)$ over $t \in [400, 460]$, blue, each panel divided by its own peak $|X|$ so the four rows are directly comparable in shape; the peak value before rescaling is printed in the panel title. The red curve is the drive $\sin(\omega t)$ at fixed amplitude 0.55, drawn as a phase reference, not to scale. **Right column:** the energy $h(t)$ over the whole run on a logarithmic vertical axis, orange; the median of h is printed in the panel title. Reference periods for this ω : drive $2\pi/\omega = 1.257$, twice the drive $4\pi/\omega = 2.513$, free period $2\pi \approx 6.283$. Settings $\varepsilon = 0.05$, $\sigma = 1$, $r = 0.5$, $T = 900$, sample interval 0.05.

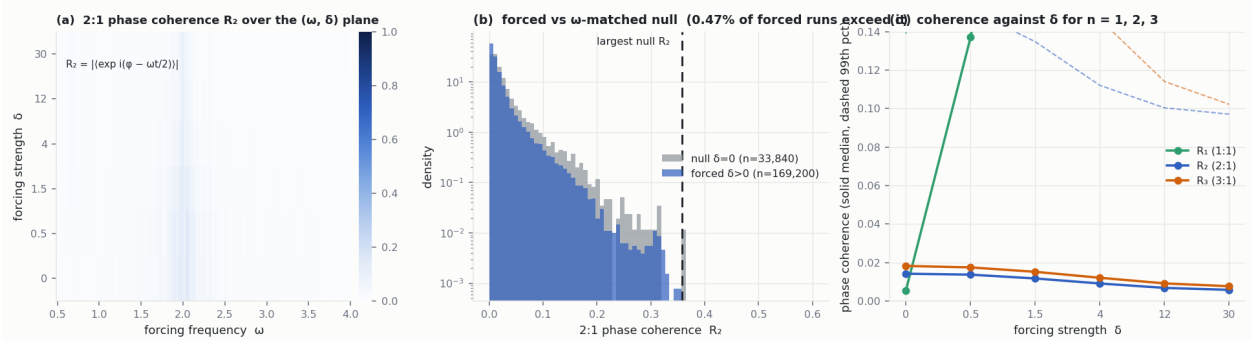
F08 — Distributions under additive forcing ($\varepsilon=0.05$, $\sigma=1$, $r=0.5$ unless stated)



F08 — distributions. (a) Stationary density of h at $\omega = 2.5$ for $\delta \in \{0, 1.5, 4, 12\}$ (coloured steps) with the closed form (2) as a black dashed curve. Horizontal axis h from 0 to 25; vertical axis density, logarithmic. (b) $\log_{10} h(t)$ over the first 400 time units, one curve per δ . (c) Histogram of the growth rate of $\log h$ over all 203,040 runs, split into $\delta = 0$ (grey) and $\delta > 0$ (blue); vertical axis is a count on a logarithmic scale, black dashed line at zero. (d) Cycle-period distributions at $\omega = 2.5$ from the phase-winding definition, for $\delta = 0$ (grey), 4 (blue), 30 (orange), pooled over 8 realisations, $T = 6000$, burn-in 1500. Three reference periods are marked and labelled in the panel: green $2\pi/\omega = 2.513$ the drive period, red $4\pi/\omega = 5.027$, black dotted 2π the free period. (e) Histogram of the relative distance of each run's median period from $4\pi/\omega$, that is $|q_{50} - 4\pi/\omega|/(4\pi/\omega)$; grey $\delta = 0$, blue $\delta > 0$; vertical axis a count on a logarithmic scale. (f) Histogram of the Sarle bimodality coefficient of the period distribution; grey $\delta = 0$, green $\delta > 0$ with $r = 0$ (exactly linear), blue $\delta > 0$ with $r > 0$. Red dashed line at $5/9$, the uniform-distribution value.

Measured: 1.16% of forced runs fall within 1% of $4\pi/\omega$, against 1.16% of the $\delta = 0$ null. Bimodality coefficient above $5/9$ in 44.5% of forced runs, 31.3% of forced runs at $r = 0$, and 1.25% of null runs.

F09 — Phase coherence under additive forcing



F09 — phase coherence. (a) Map of the 2:1 coherence $R_2 = |\langle \exp i(\varphi - \omega t/2) \rangle|$. Horizontal axis ω ; vertical axis forcing strength δ as a categorical axis with the six sweep values as ticks; colour R_2 on a fixed scale from 0 to 1. Slice shown: $\varepsilon = 0.05$, $\sigma = 1$, $r = 0.5$, averaged over seeds. (b) Densities of R_2 for the 33,840 unforced runs (grey) and the 169,200 forced runs (blue); vertical axis logarithmic. The black dashed vertical line is the single largest R_2 observed anywhere in the null. The panel title reports the fraction of forced runs exceeding their ω -matched null, that is the null maximum taken among runs with the same $|\omega/n - 1|$, rather than the pooled maximum. (c) Coherence against δ (categorical axis): solid lines the median, dashed lines the 99th percentile, for R_1 (green), R_2 (blue), R_3 (orange).

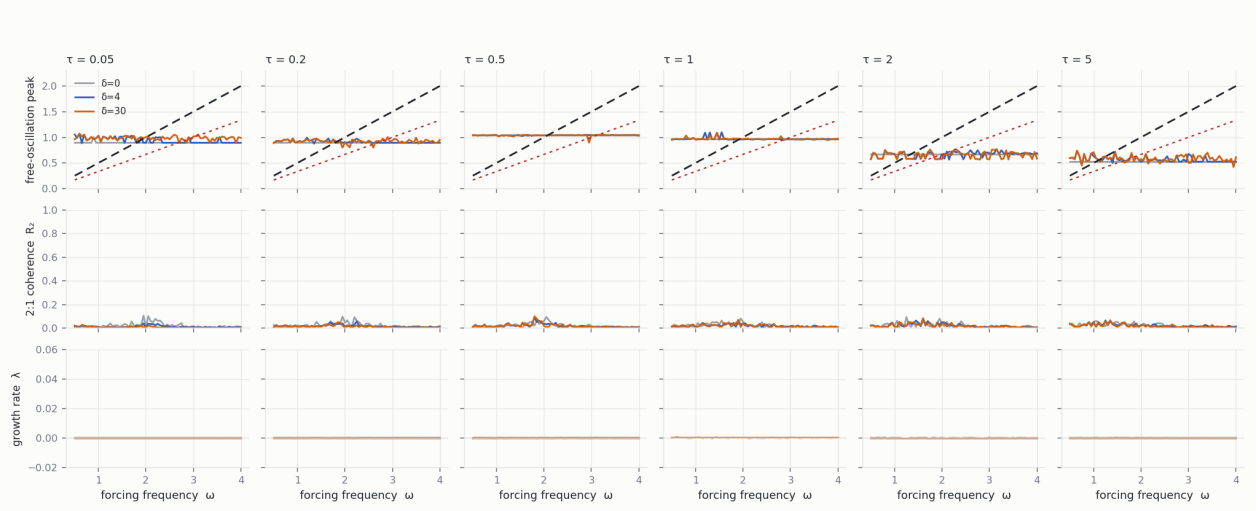
Measured: largest null $R_2 = 0.3574$; median R_2 is 0.0092 for forced runs against 0.0142 for the null; 0.47% of forced runs exceed their ω -matched null in R_2 and 0.74% in R_3 .

4 The full three-dimensional system

Simulation S2 — three-dimensional scan (figure). $\varepsilon = 0.2$, $\sigma = 1$; $\tau \in \{0.05, 0.2, 0.5, 1, 2, 5\}$; $\delta \in \{0, 4, 30\}$; ω from 0.5 to 4.0 in steps of 0.05 (71 values); $T = 6000$, burn-in 2000. Here $\varepsilon = 0.2$ is used rather than 0.05 because it keeps the random frequency's own parametric instability weak enough that the growth rate stays measurable.

Simulation S3 — three-dimensional scan (tabulated). $\varepsilon \in \{0.2, 0.5\}$, $\tau \in \{0.5, 1, 2, 5\}$, $\delta \in \{0, 1, 4, 12, 30\}$, ω from 1.0 to 4.0 in steps of 0.05 (61 values), seeds $\{11, 22, 33, 44\}$: **9,760** runs, stored in `scan3d_full.csv` with the sub-drive peak, R_2 and R_3 for each.

F10 — The full three-dimensional system with additive forcing, $\varepsilon=0.2$, $\sigma=1$. τ is the correlation time of the random frequency A ; the reduced two-dimensional model is the $\tau \rightarrow 0$ limit. Columns are values of τ . Rows: free-oscillation spectral peak (drive harmonics excluded), 2:1 coherence R_2 , growth rate λ of $\log h$. Dashed guides in the top row: black $\omega/2$, red $\omega/3$.



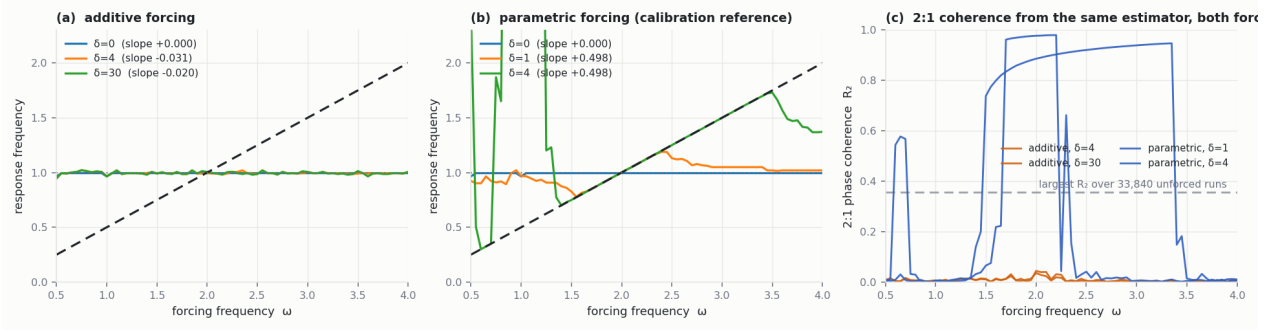
F10 — the full three-dimensional system under additive forcing. Columns are values of the correlation time τ , increasing left to right: 0.05, 0.2, 0.5, 1, 2, 5. All three rows share the horizontal axis, forcing frequency ω from 0.5 to 4.0. In every panel the three curves are grey $\delta = 0$, blue $\delta = 4$, orange $\delta = 30$. **Top row:** the free-oscillation spectral peak of X , computed with the four drive harmonics excised, vertical axis 0 to 2.3; the black dashed guide is $\omega/2$ and the red dotted guide is $\omega/3$. **Middle row:** 2:1 coherence R_2 , vertical axis 0 to 1. **Bottom row:** growth rate λ of $\log h$, vertical axis -0.02 to 0.06 , with a light line at zero.

Measured at $\delta = 30$, slope of the free peak over $\omega \in [1.8, 2.2]$: $+0.005$, $+0.023$, -0.001 , -0.030 , $+0.144$, -0.173 for $\tau = 0.05, \dots, 5$; the mean R_2 over $\omega \in (1.6, 2.6)$ is 0.009 to 0.032 across the same range of τ .

5 Calibration of the detector

Simulation S4. The identical measurement pipeline — same integrator, same run length $T = 6000$ and burn-in 2000, same sub-drive peak finder, same fixed-window regression on $\omega \in [1.8, 2.2]$, same coherence estimator — is applied to two forcings at $\varepsilon = 0.05$, $\sigma = 1$, $r = 0.5$, ω from 0.5 to 4.0 in steps of 0.05: additive at $\delta \in \{0, 4, 30\}$ and, *purely as a calibration reference*, parametric at $\delta \in \{0, 1, 4\}$. This is the only place in this document where a parametric run appears, and it appears only to establish what the detector does when the response does follow the drive.

F10b — Detector calibration: the identical measurement pipeline (integrator, run length, peak finder, fixed-window regression, coherence estimator) applied to additive and to parametric forcing.



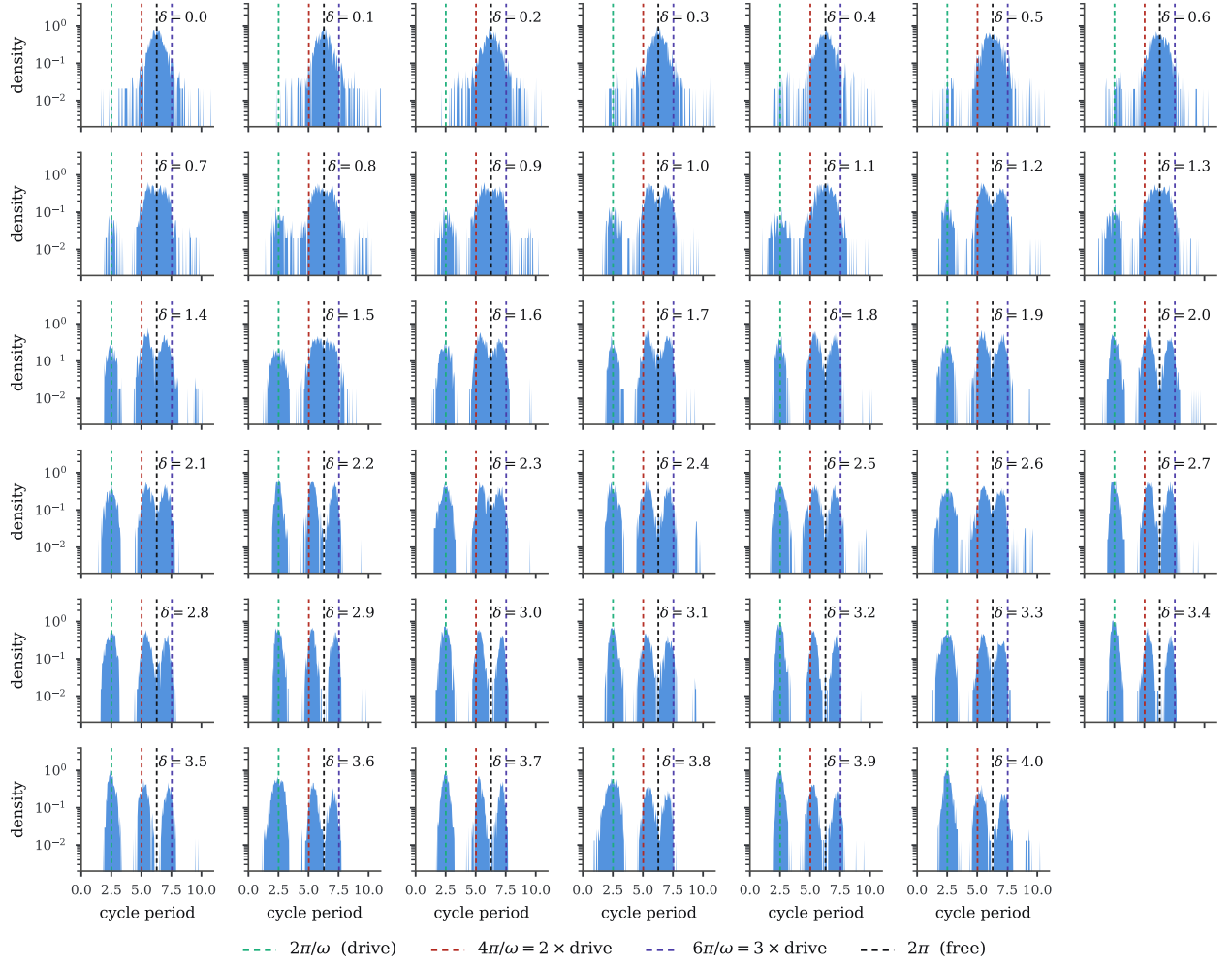
F10b — detector calibration. (a) Additive forcing. Horizontal axis ω ; vertical axis the sub-drive response frequency, 0 to 2.3. One curve per δ , with its fitted tracking slope printed in the legend. The black dashed guide is $\omega/2$; the grey dotted horizontal line is $\omega_0 = 1$. (b) The same axes and the same estimator applied to parametric forcing, included as the calibration reference. (c) 2:1 coherence R_2 from the same estimator, both forcings on one pair of axes: orange additive, blue parametric. The grey dashed horizontal line at 0.3574 is the largest R_2 seen anywhere among the 33,840 unforced runs of Simulation S1.

Measured tracking slope over $\omega \in [1.8, 2.2]$: additive -0.031 at $\delta = 4$ and -0.020 at $\delta = 30$; parametric $+0.498$ at both $\delta = 1$ and $\delta = 4$. Mean R_2 over the same window: additive 0.022 and 0.016; parametric 0.977 and 0.884.

6 Cycle length under the phase-winding definition

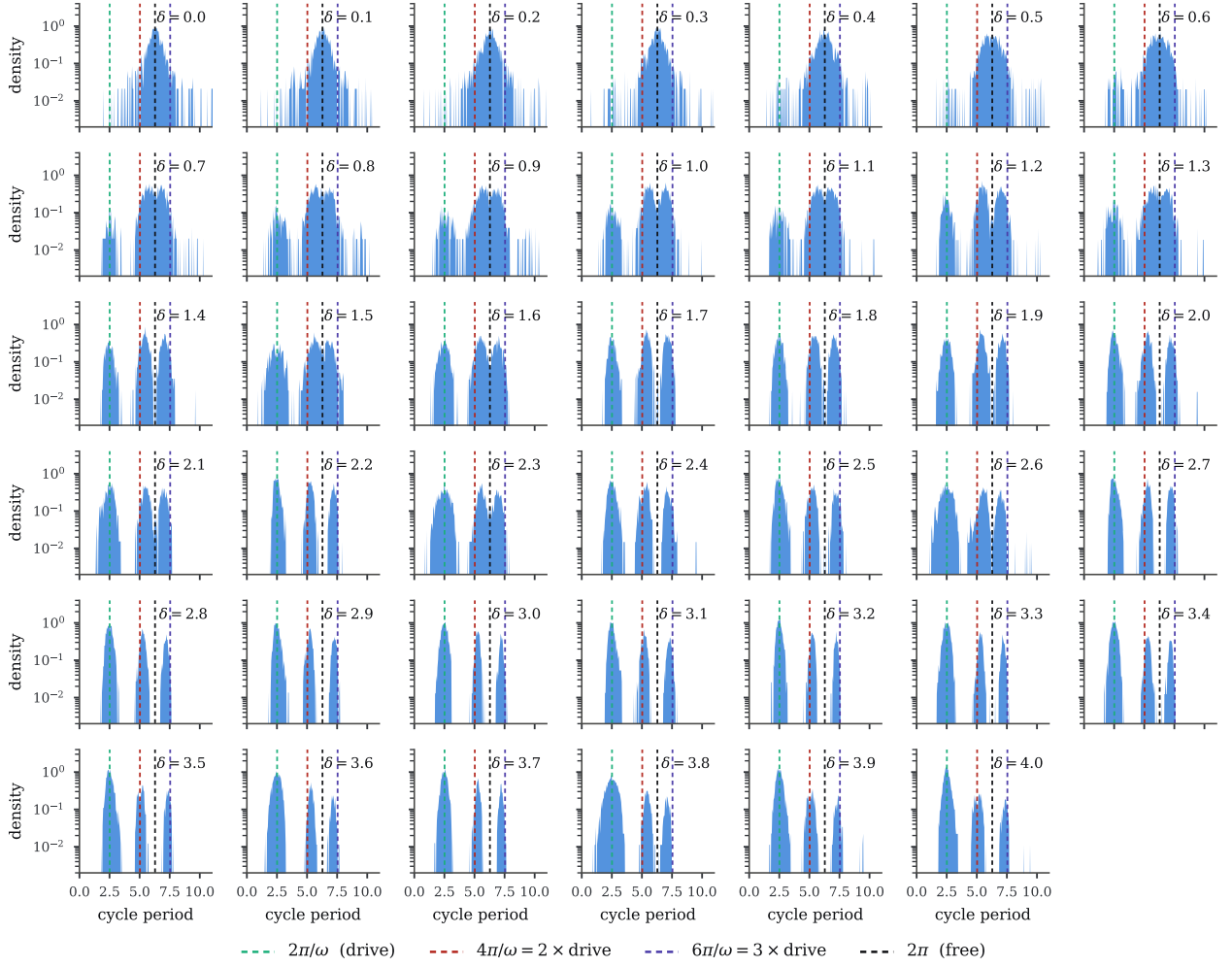
Simulation S5 — the δ sweep at $\omega = 2.5$. $\varepsilon = 0.05$, $\sigma = 1$, $\omega = 2.5$ fixed; δ from 0 to 4 in steps of 0.1 (41 values); 12 realisations per δ ; $T = 8000$, burn-in 2000. Run twice, at $r = 0.5$ and at $r = 0$. Cycle periods are histogrammed on 280 bins over $[0, 14]$ and stored, so the later figures need no re-simulation. Output: `psweep_r05.npz`, `psweep_r00.npz`. At this ω the reference periods are $2\pi/\omega = 2.513$, $4\pi/\omega = 5.027$, $6\pi/\omega = 7.540$ and $2\pi = 6.283$.

Cycle-period distribution as δ sweeps 0 to 4, additive forcing, $\omega = 2.5$, $r = 0.5$
(log density, so the small modes are visible)

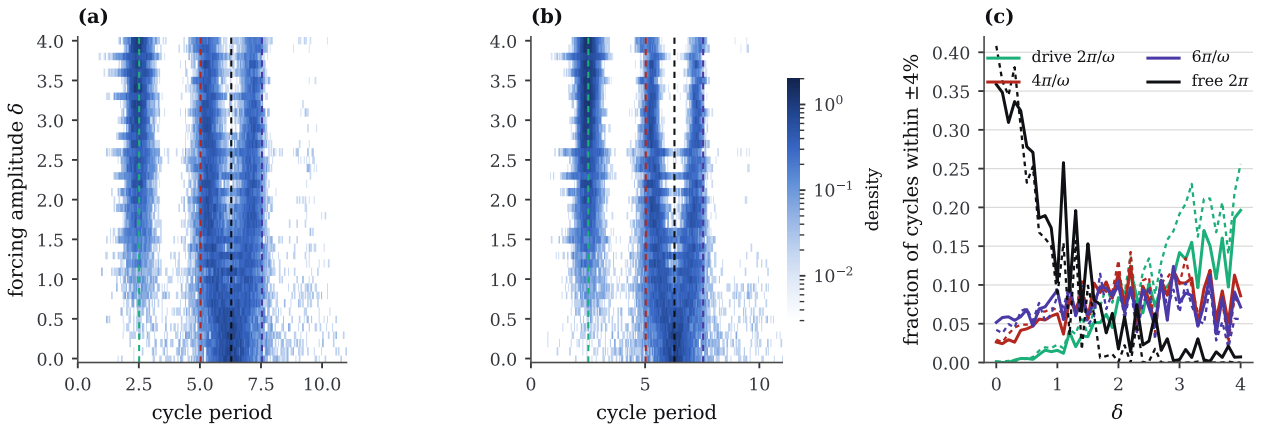


F19 — cycle-period distribution as δ sweeps $0 \rightarrow 4$, $r = 0.5$. Forty-one small panels, one per value of δ , reading left to right and top to bottom from $\delta = 0$ to $\delta = 4$; the value of δ is printed in the top right of each panel. In each panel the horizontal axis is the cycle period from 0 to 11 and the vertical axis is probability density on a *logarithmic* scale from 2×10^{-3} to 4, chosen so that small secondary modes remain visible. The blue filled area is the measured distribution. The four dashed vertical lines are, in the colours given in the legend below the grid: green $2\pi/\omega$, the drive period; red $4\pi/\omega$, twice the drive period; purple $6\pi/\omega$, three times the drive period; black 2π , the free period.

Cycle-period distribution as δ sweeps 0 to 4, additive forcing, $\omega = 2.5$, $r = 0$ (exactly linear)
(log density, so the small modes are visible)

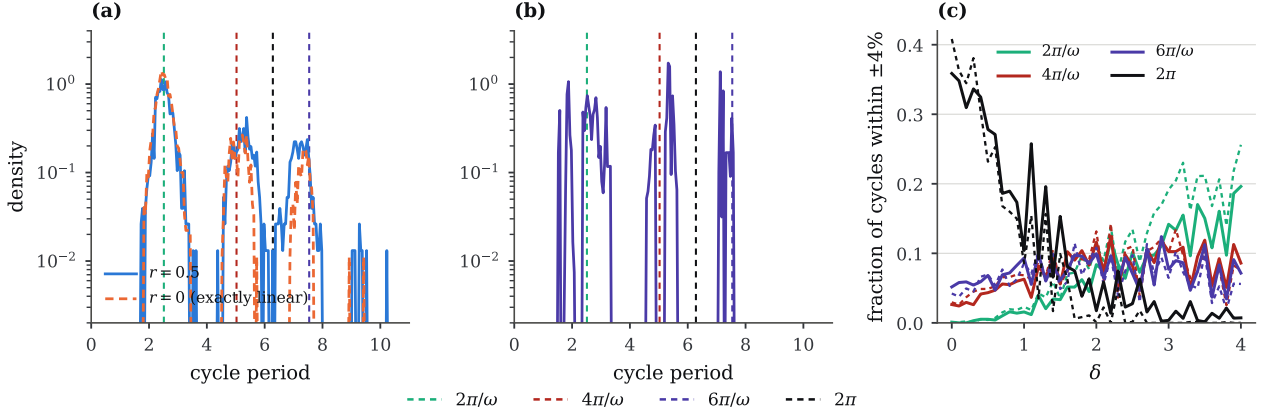


F19b — the same sweep at $r = 0$, the exactly linear control. Identical layout, axes, scales and reference lines to F19; the only change is $r = 0$, for which the system is exactly linear.



F19c — the same data as maps and as band fractions. (a) and (b): the whole sweep as a single map, for $r = 0.5$ and $r = 0$ respectively. Horizontal axis: cycle period, 0 to 11. Vertical axis: forcing amplitude δ , 0 to 4. Colour: probability density on a logarithmic scale shared by both panels, 3×10^{-3} to 2. Each horizontal row of a map is one panel of F19 laid flat. The four vertical dashed lines are the same reference periods as in F19. (c) The fraction of all measured cycles falling within $\pm 4\%$ of each reference period, plotted against δ . Colours match the reference lines: green $2\pi/\omega$, red $4\pi/\omega$, purple $6\pi/\omega$, black 2π . Solid curves $r = 0.5$, dashed curves $r = 0$.

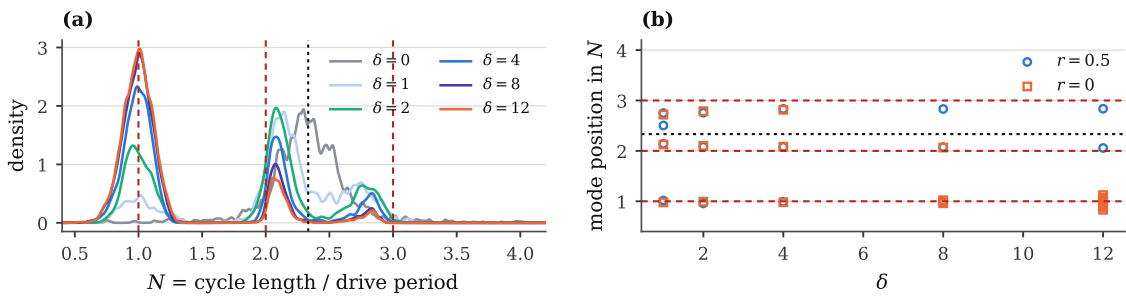
Simulation S6 — the two-tone null. A pure superposition $X(t) = \sin(\omega_0 t + \phi) + B \sin(\omega t)$ with $\omega_0 = 1$, $\omega = 2.5$, random phase ϕ , 40 realisations, sampled at $\Delta t = 0.2$ over $t \in [0, 40000]$. The amplitude ratio B is set to the linear-response value at $\delta = 4$, giving $B/A = 0.76$. This object has no dynamics at all: its cycle-period distribution is pure two-tone geometry, measured with the same phase-winding routine as the simulations.



F20 — simulation against the two-tone null. (a) Cycle-period distribution at $\delta = 4$ from Simulation S5. Horizontal axis cycle period 0 to 11; vertical axis density, logarithmic, 2×10^{-3} to 4. Blue solid $r = 0.5$; orange dashed $r = 0$. (b) The same axes and scales, for the two-tone null of Simulation S6 at $B/A = 0.76$. (c) Fraction of cycles within $\pm 4\%$ of each reference period against δ , solid $r = 0.5$ and dashed $r = 0$, as in F19c(c). In all three panels the vertical dashed reference lines are green $2\pi/\omega$, red $4\pi/\omega$, purple $6\pi/\omega$, black 2π .

Simulation S7 — modes at the QBO-analogue frequency. $\omega = 2.3333$, chosen because the free period then contains 2.333 drive periods, the same ratio as a 28-month cycle to an annual one. $\varepsilon = 0.05$, $\sigma = 1$; $\delta \in \{0, 1, 2, 4, 8, 12\}$; $r \in \{0.5, 0\}$; 24 realisations; $T = 12000$, burn-in 3000. Cycle lengths are converted to N and the modes of the distribution are located by kernel density estimation (bandwidth factor 0.035) followed by peak detection with a prominence threshold of 2–3% of the maximum. Output: modes.npz.

Cycle length in drive periods, $N = T_{\text{cycle}}/T_{\text{drive}}$, phase-winding definition.
Red dashed: integers 1, 2, 3. Black dotted: $N = \omega = 2.333$, the unforced value.



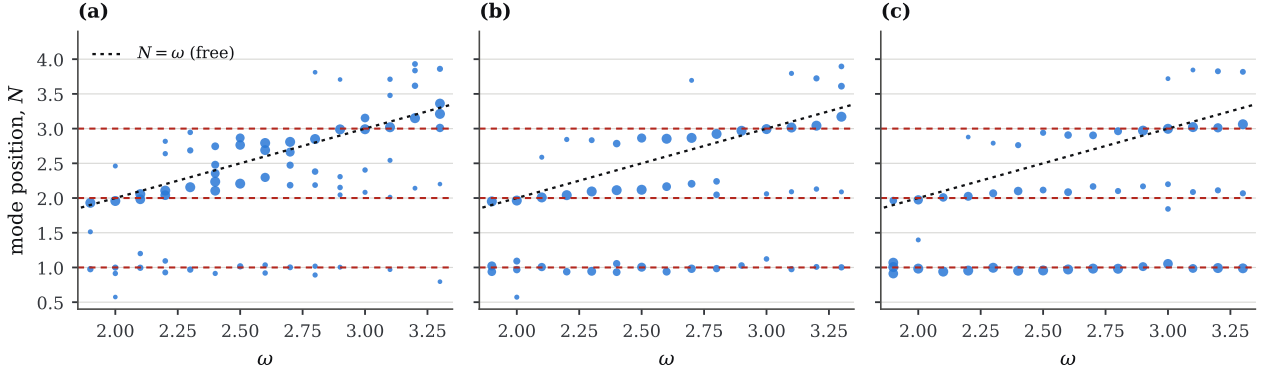
F21 — cycle length in drive periods at $\omega = 2.333$. (a) Kernel density estimates of $N = T_{\text{cycle}}/T_{\text{drive}}$ at $r = 0.5$. Horizontal axis N from 0.4 to 4.2; vertical axis density. One curve per forcing strength, from grey $\delta = 0$ through to orange $\delta = 12$ as given in the legend. Red dashed vertical lines mark the integers $N = 1, 2, 3$; the black dotted line marks $N = \omega = 2.333$, the value the unforced system takes exactly. (b) The positions of every detected mode, plotted against δ . Horizontal axis δ ; vertical axis mode position in N , same range as (a). Blue circles $r = 0.5$, orange squares $r = 0$. The same red dashed integer lines and black dotted $N = \omega$ line are drawn horizontally.

Measured mode weights within ± 0.04 of each target, at $r = 0.5$: at $\delta = 0$, 0.28% at $N = 1$, 18.1% at $N = 2$, 2.5% at $N = 3$ and 47.7% at $N = \omega$; at $\delta = 2$, 30.3%, 31.6%, 5.1% and 9.6%; at $\delta = 4$, 53.3%, 21.2%, 2.7% and 3.8%.

Simulation S8 — every mode against ω . $\varepsilon = 0.05$, $\sigma = 1$, $r = 0.5$; ω from 1.90 to 3.30 in steps of 0.10 (15 values); $\delta \in \{1, 2, 4\}$; 12 realisations per cell; $T = 6000$, burn-in 1500. Modes are found as in S7 with

a prominence threshold of 5%. Output: `allpeaks.npz`. The same run was repeated at $r = 0$ for $\delta \in \{2, 4\}$ (`allpeaks_r0.npz`).

Every detected mode of the cycle-length distribution, additive forcing. Marker size = relative height.



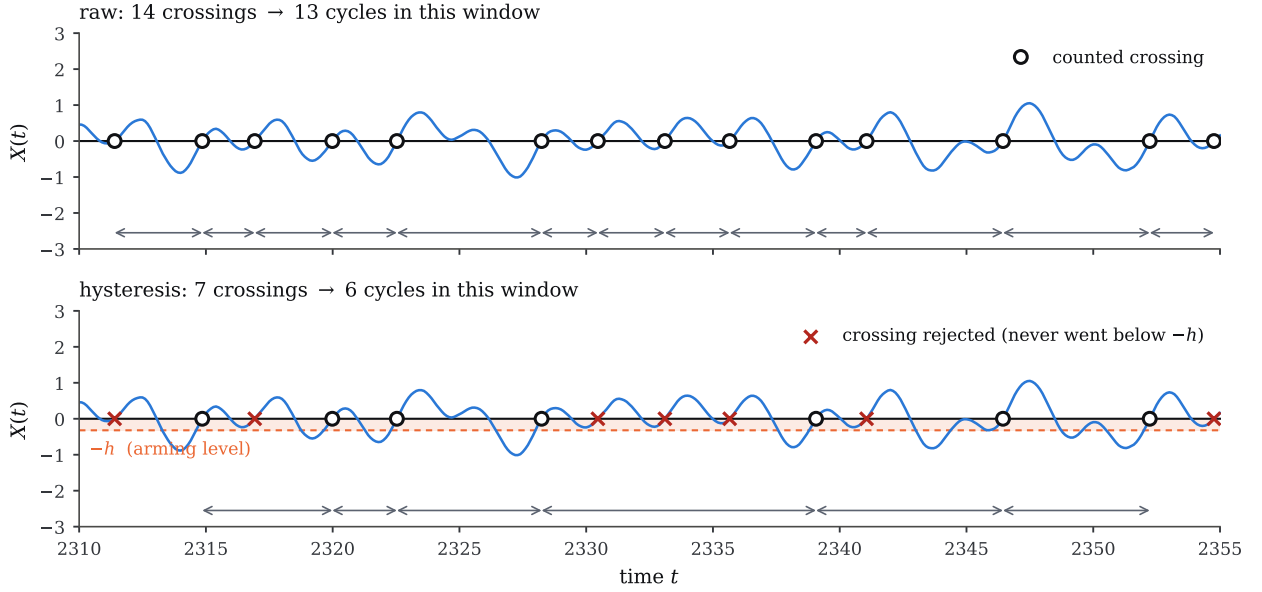
F22 — every detected mode of the cycle-length distribution, phase-winding definition. Three panels, $\delta = 1, 2, 4$ left to right. Horizontal axis: forcing frequency ω , 1.9 to 3.3. Vertical axis: the position in N of a detected mode. *Every dot is one mode of one distribution*: for a given ω the cycle-length distribution is estimated, its local maxima are found, and one dot is plotted at each maximum. A column of dots at one ω therefore shows all the modes that distribution has, not a scatter of individual cycles. The area of a dot is proportional to the height of that mode relative to the tallest mode in the same column, so small dots are minor modes and large dots are dominant ones. Red dashed horizontal lines: the integers $N = 1, 2, 3$. Black dotted diagonal: $N = \omega$, the value the unforced system takes.

7 Cycle length under the zero-crossing definitions

7.1 The two counting rules

An upward zero crossing of X is located by linear interpolation between the two samples that bracket the sign change. The *raw* rule counts every one. The *hysteresis* rule adds a Schmitt trigger: after an accepted crossing, the signal must fall below $-h_{\text{th}}$ before another upward crossing can be accepted, with $h_{\text{th}} = 0.25 \text{rms}(X)$ computed from the run itself. Small wiggles that carry X back and forth across zero without a full excursion are therefore counted by the raw rule and discarded by the hysteresis rule.

The same trajectory read two ways. $\omega = 2.333$, $\delta = 2$, $h = 0.25$ $\text{rms}(X) = 0.32$.



F23 — the same trajectory read by the two rules. Both panels show the identical segment of one run, $\omega = 2.333$, $\delta = 2$, over $t \in [2310, 2355]$. Horizontal axis time; vertical axis $X(t)$, blue. **Top:** the raw rule. Open circles on the zero line: every counted crossing. Grey double-headed arrows beneath the trace: the resulting cycles, one arrow per cycle. **Bottom:** the hysteresis rule. The orange shaded band between 0 and $-h_{\text{th}}$ and the orange dashed line at $-h_{\text{th}}$ show the arming level the signal must pass below to re-arm the trigger. Open circles: crossings that were accepted. Red crosses: crossings that the raw rule counted and this rule rejected because X never went below $-h_{\text{th}}$ since the previous accepted crossing. Each panel title gives the number of crossings and the number of cycles in the window.

Measured in this window: $h_{\text{th}} = 0.55$; the raw rule finds 14 crossings, the hysteresis rule accepts 7 and rejects 7.

7.2 The fine sweep

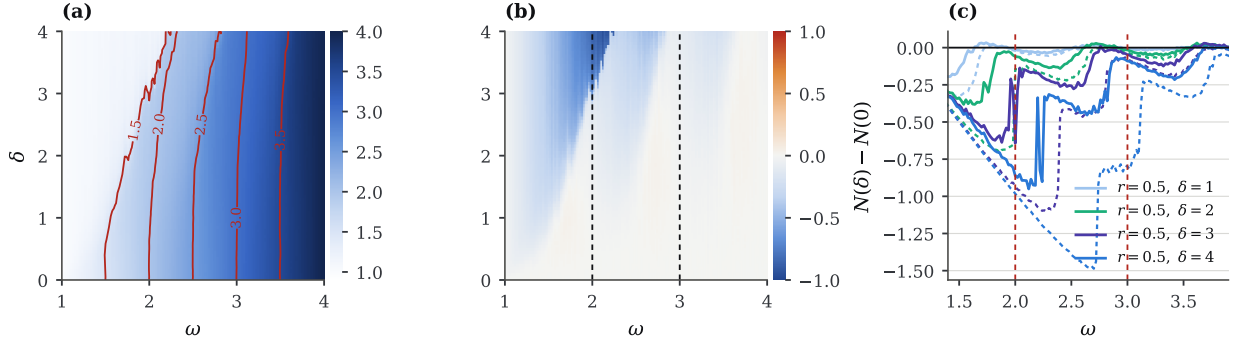
Simulation S9 — the fine (ω, δ) zero-crossing sweep. The grid is

$$\omega = 1.00, 1.02, \dots, 4.00 \text{ (151 values)}, \quad \delta = 0.00, 0.05, \dots, 4.00 \text{ (81 values)},$$

giving **12,231** cells, each run with three seeds $\{11, 22, 33\}$ and pooled. Settings $\varepsilon = 0.05$, $\sigma = 1$, $T = 4000$, $\Delta t = 0.0025$, burn-in 1000, sample interval 0.2, 400 configurations per integrator call. The whole grid is run twice, at $r = 0.5$ and at $r = 0$. For every cell, and separately for each of the two counting rules, the full histogram of N is stored on 240 bins over $[0, 6]$ together with the median and the cycle count, so every figure below is made from stored histograms without re-simulating. Output: `zc_r05.npz`, `zc_r00.npz`.

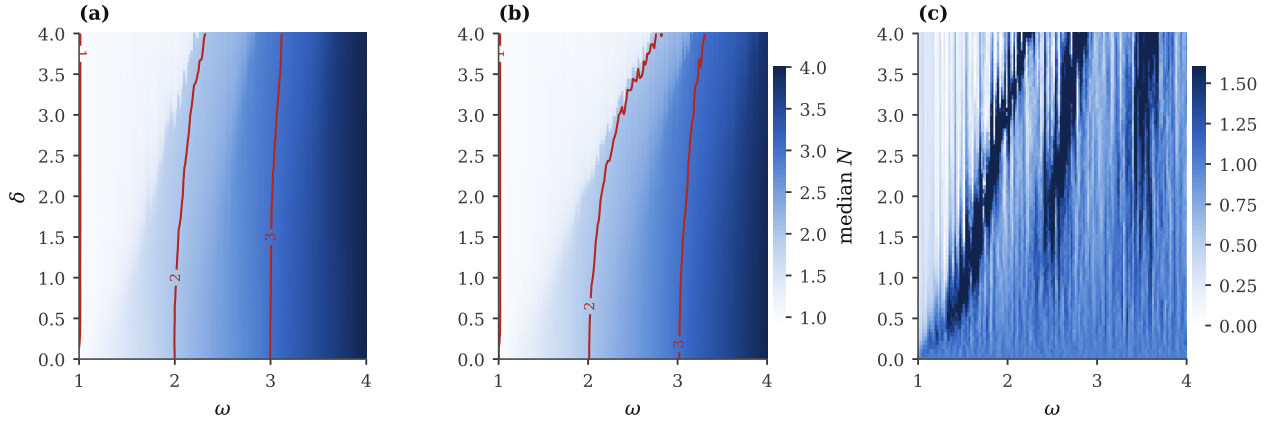
Sample size and validation. The median cell contains **1,480** cycles and the smallest contains 1,342. At $\delta = 0$ the measured median N should equal ω exactly; the measured values are 1.5017, 2.0023, 2.5029, 3.0034 and 3.5040 at $\omega = 1.5, 2.0, 2.5, 3.0, 3.5$, an offset of $+0.0017$ to $+0.0040$ attributable to finite sampling of the crossing times. The two values of r give median N surfaces whose typical absolute difference is 0.030.

Cycle length from zero crossings of X (hysteresis), $N = T_{\text{cycle}}/T_{\text{drive}}$. 12,231 cells, 3 seeds each.



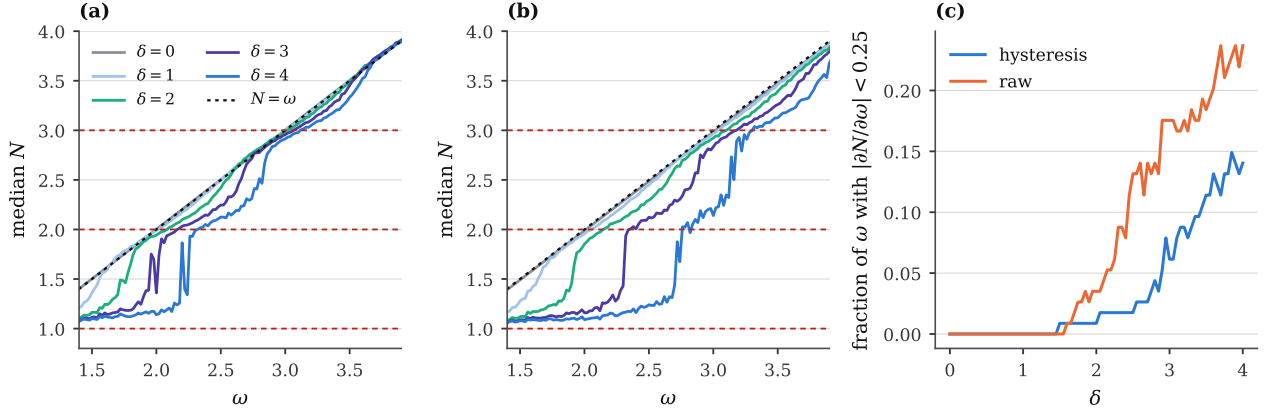
F24 — the median cycle length over the (ω, δ) plane. All three panels use the hysteresis rule at $r = 0.5$. **(a)** Map of the median N . Horizontal axis ω from 1 to 4; vertical axis δ from 0 to 4; colour median N from 0.9 (pale) to 4.0 (dark). Red contours are drawn and labelled at median $N = 1.5, 2, 2.5, 3, 3.5$. **(b)** The same plane showing the *change* caused by the forcing, $N(\delta) - N(0)$, so each column is measured against its own unforced value at the same ω . Colour is diverging and symmetric about zero, from -1 to $+1$; black dashed vertical lines mark $\omega = 2$ and $\omega = 3$. **(c)** Cuts through panel (b) at four forcing strengths. Horizontal axis ω from 1.4 to 3.9; vertical axis $N(\delta) - N(0)$. Solid curves $r = 0.5$, dashed curves $r = 0$, one colour per $\delta \in \{1, 2, 3, 4\}$; the horizontal line is zero and red dashed verticals mark $\omega = 2, 3$.

Measured: the fraction of ω values at which $N(\delta) - N(0)$ is negative is 57.3%, 75.0%, 87.1%, 94.4% at $\delta = 1, 2, 3, 4$ for $r = 0.5$, and 64.5%, 86.3%, 96.8%, 100% for $r = 0$. The corresponding means of $N(\delta) - N(0)$ are $-0.013, -0.082, -0.196, -0.358$ at $r = 0.5$ and $-0.038, -0.196, -0.441, -0.740$ at $r = 0$.



F24b — the same surface under both counting rules, with its gradient. **(a)** Median N over the (ω, δ) plane, hysteresis rule. **(b)** The same, raw rule. Both use the identical colour scale, 0.9 to 4.0, with red contours labelled at median $N = 1, 2, 3$. **(c)** The magnitude of the slope of the surface in (a) along the ω direction, $|\partial N / \partial \omega|$, computed by finite differences on the ω grid. Colour runs from 0 to 1.6. On this panel a value of 0 means the median cycle length does not change as ω is varied at that point, and a value of 1 means it changes as fast as ω itself.

Measured: the median- $N = 2$ contour sits at $\omega = 1.998$ at $\delta = 0$ and moves to $\omega = 2.014, 2.083, 2.162, 2.318$ at $\delta = 1, 2, 3, 4$.



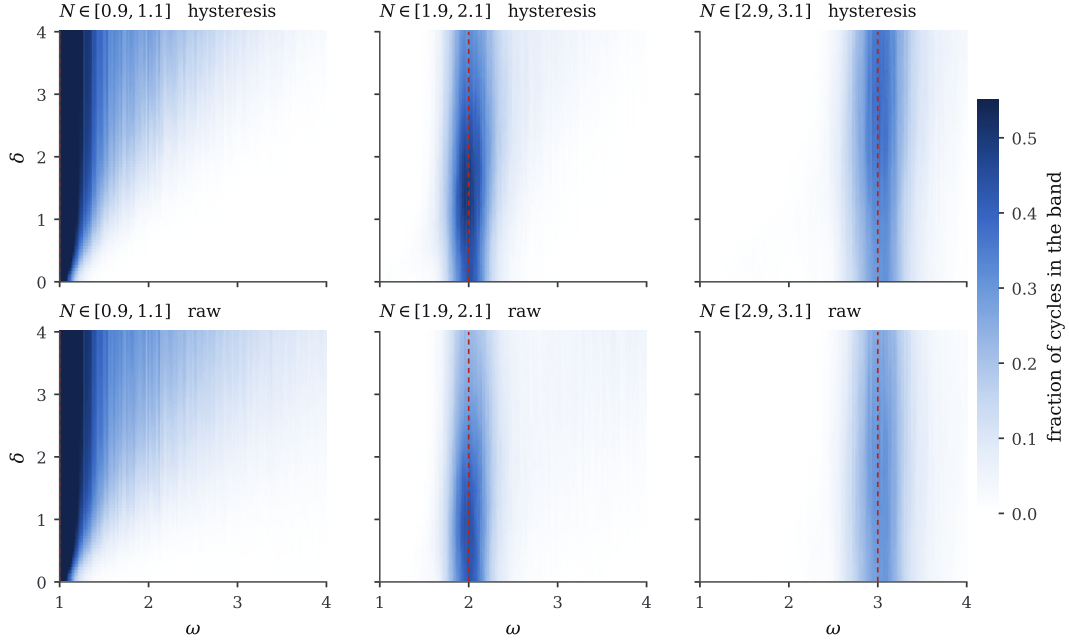
F25 — median N against ω , and how much of the range is flat. (a) Median N against ω at five forcing strengths, hysteresis rule. Horizontal axis ω from 1.4 to 3.9; vertical axis median N from 0.8 to 4.0. One curve per $\delta \in \{0, 1, 2, 3, 4\}$, grey to blue as given in the legend. The black dotted diagonal is $N = \omega$, the unforced relation. Red dashed horizontals mark the integers 1, 2, 3. (b) The identical plot for the raw rule; same axes, same curves, same guides. (c) Plateau coverage: the fraction of $\omega \in (1.5, 3.8)$ at which $|\partial N / \partial \omega| < 0.25$, plotted against δ . Blue hysteresis, orange raw. A value of 0 means the curve in (a) or (b) is nowhere flatter than a quarter of the unforced slope; a value of 1 would mean it is flat everywhere.

Measured plateau coverage at $r = 0.5$, hysteresis: 0 at $\delta = 0, 1, 2$; 8.1% at $\delta = 3$; 19.4% at $\delta = 4$.

7.3 Occupancy of the integer bands

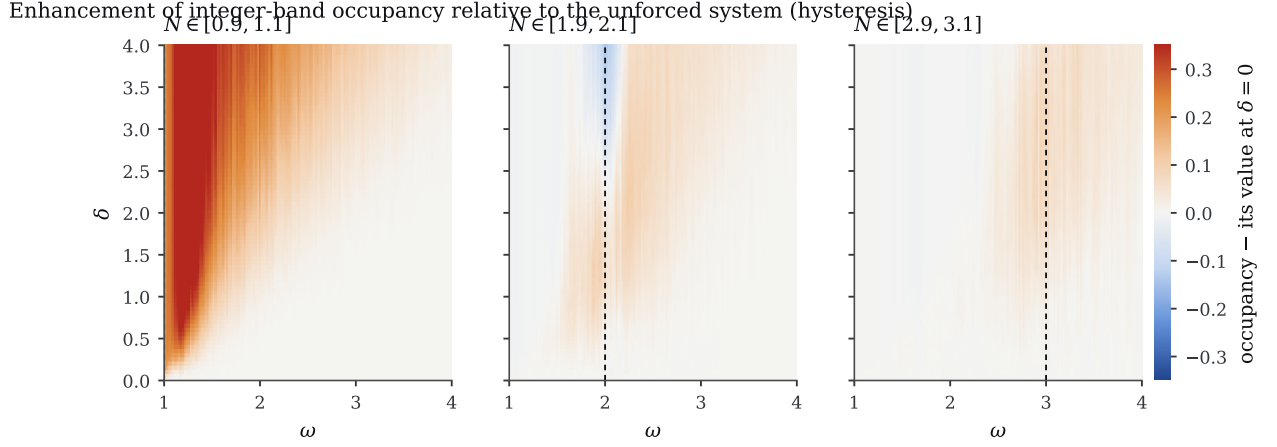
For each cell, the fraction of measured cycles with N inside $[0.9, 1.1]$, $[1.9, 2.1]$ or $[2.9, 3.1]$ is computed directly from the stored histogram. These are the “integer bands”.

Occupancy of the integer bands over the (ω, δ) plane. Dashed line: $\omega = N$, where the unforced system already sits in the band.



F27 — occupancy of the integer bands over the (ω, δ) plane. Six maps. The top row uses the hysteresis rule, the bottom row the raw rule; columns are the bands $N \in [0.9, 1.1]$, $[1.9, 2.1]$ and $[2.9, 3.1]$ from left to right. In every map the horizontal axis is ω from 1 to 4, the vertical axis is δ from 0 to 4, and the colour is the fraction of that cell’s cycles lying inside the band, on a scale shared by all six panels from 0 to 0.55. The red dashed vertical line in each panel is at ω equal to the band’s integer, which is the ω at which the unforced system already has N inside that band.

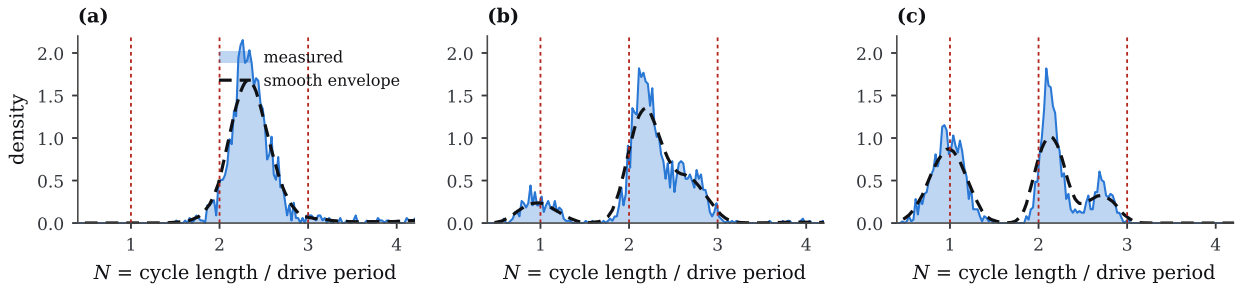
Measured at $\delta = 4$ (hysteresis), the largest occupancy of each band and the value at the same ω when $\delta = 0$: band 1, 0.933 at $\omega = 1.00$ against 0.709; band 2, 0.321 at $\omega = 1.96$ against 0.415; band 3, 0.364 at $\omega = 3.02$ against 0.301.



F26 — the same occupancy measured against its own unforced value. Three maps, one per band, hysteresis rule. Axes as in F27: horizontal ω from 1 to 4, vertical δ from 0 to 4. The quantity plotted is the band occupancy *minus* the occupancy of the same band at $\delta = 0$ and the same ω , so each column is referenced to its own unforced value and the whole $\delta = 0$ row is zero by construction. The colour scale is diverging and symmetric, -0.35 to $+0.35$: one colour for occupancy above the unforced value, the other for below, and the neutral midpoint for no change. The black dashed vertical line marks ω equal to the band's integer.

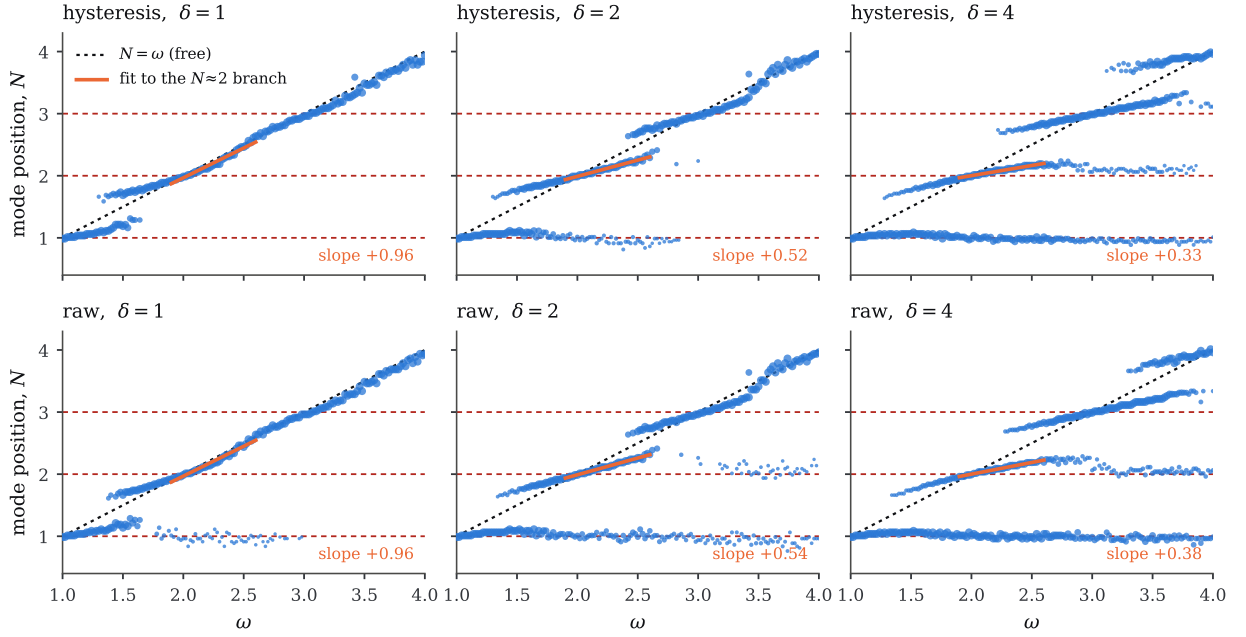
7.4 Shape of the distribution at a fixed frequency

Cycle-length distribution at $\omega = 2.333$, with a wide-bandwidth smooth envelope. A "lump" is the measured curve rising ABOVE the dashed envelope at an integer.



F28 — the distribution of N at $\omega = 2.333$ against a smooth envelope. Three panels, $\delta = 0, 2, 4$ left to right, hysteresis rule, $r = 0.5$, taken from the stored histograms of Simulation S9. Horizontal axis: N from 0.4 to 4.2. Vertical axis: probability density, linear, 0 to 2.2. The blue shaded area with its thin blue outline is the measured distribution. The black dashed curve is the same distribution convolved with a wide Gaussian kernel (full width at half maximum 0.40 in N), which removes any structure narrower than that width while preserving the broad shape — it is the distribution the data would have if it had no narrow features. Red dotted vertical lines mark the integers $N = 1, 2, 3$. Reading the panel means comparing the solid curve to the dashed one at those three positions: where the measured density rises above its own smooth envelope at an integer, there is narrow excess mass there; where it sits on or below the envelope, there is none.

Every detected mode of the cycle-length distribution, additive forcing, zero-crossing measure. Marker size = height relative to the tallest mode in that column. ω step 0.02.



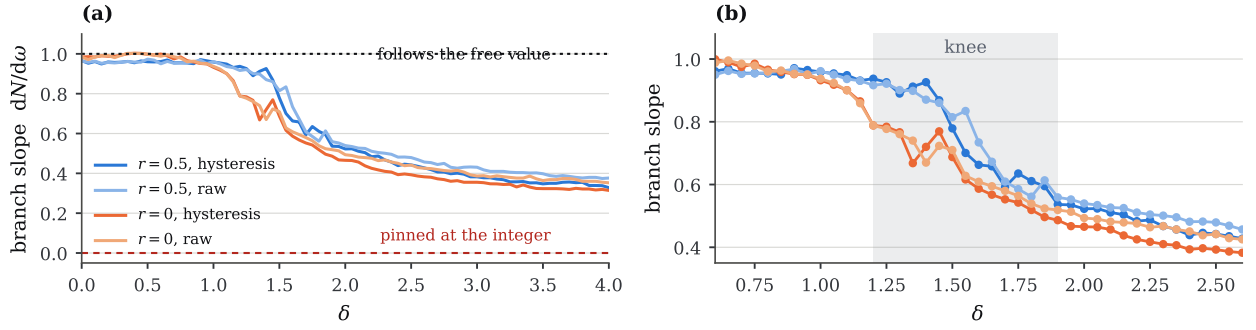
F29 — every detected mode, zero-crossing definition. The zero-crossing counterpart of F22, built from the stored histograms of Simulation S9, so the ω grid has step 0.02 (151 values) rather than the step 0.10 (15 values) of the phase-winding version, and branches appear as near-continuous curves rather than isolated columns. Six panels: the top row is the hysteresis rule and the bottom row the raw rule; columns are $\delta = 1, 2, 4$. Horizontal axis ω from 1 to 4; vertical axis mode position in N , 0.4 to 4.3. As in F22, every dot is one mode of one cell's distribution, and the marker area is proportional to that mode's height relative to the tallest mode in the same column. Black dotted diagonal: $N = \omega$. Red dashed horizontals: the integers 1, 2, 3. The thick orange segment in each panel is a straight-line fit to the dots belonging to the branch that begins near $N = 2$, restricted to $\omega \in [1.9, 2.6]$ and $N \in (1.6, 2.6)$; its slope is printed in the corner of the panel.

Measured slope of the $N \approx 2$ branch: hysteresis +0.959, +0.523, +0.328 at $\delta = 1, 2, 4$; raw +0.961, +0.540, +0.376. For the branch near $N = 1$: hysteresis slope +0.424, -0.081 , -0.040 with mean N of 1.090, 1.004, 0.988; raw slope -0.087 , -0.051 , -0.031 with mean N of 1.015, 0.994, 1.001. Total dots per panel range from 218 to 412, with a median of 1–3 modes per ω column.

7.5 Sensitivity of the branch to the drive

Derived quantity S10. For every δ on the sweep grid (0 to 4 in steps of 0.05, 81 values), the slope $dN/d\omega$ of the branch near $N = 2$ is estimated exactly as in F29 and recorded, for all four combinations of $r \in \{0.5, 0\}$ and the two counting rules. Output: `slope_vs_delta.npz`. This is a re-analysis of the stored histograms of Simulation S9; no new simulation was run.

Sensitivity of the observed cycle length to the drive, against forcing amplitude. Slope 1 = the forcing does nothing; slope 0 = the cycle length is pinned.



F30 — branch slope against forcing amplitude. (a) The whole range. Horizontal axis δ from 0 to 4; vertical axis the fitted branch slope $dN/d\omega$, from -0.05 to 1.1 . Four curves, one per combination of r and counting rule, as given in the legend. Two horizontal guides label the ends of the scale: a slope of 1, at which the measured cycle length moves with ω exactly as the unforced system's does, and a slope of 0, at which the measured cycle length does not move with ω at all. (b) The same four curves with markers on every grid point, zoomed to $\delta \in [0.6, 2.6]$ and slope $\in [0.35, 1.08]$. The shaded vertical band spans $\delta \in [1.2, 1.9]$ and is labelled "knee".

Measured: the plateau value at small δ is 0.9555, 0.9538, 0.9568, 0.9559 for the four curves. The departure from that plateau begins at $\delta = 1.50$ for $r = 0.5$ and $\delta = 1.20$ for $r = 0$. The slope at $\delta = 4$ is 0.328 and 0.376 ($r = 0.5$, hysteresis and raw) and 0.314 and 0.353 ($r = 0$). The rate of fall $d(\text{slope})/d\delta$ in three windows is: over $\delta \in [0.2, 1.1]$, -0.004 , -0.010 , -0.093 , -0.089 ; over $\delta \in [1.4, 2.0]$, -0.605 , -0.620 , -0.480 , -0.369 ; over $\delta \in [2.2, 4.0]$, -0.078 , -0.074 , -0.058 , -0.062 .

8 Index of simulations

ID	What was run	Grid	Runs	Output
V1	Integrator validation, unforced and short forced runs	$\varepsilon=0.05$, $\sigma=1$, $r=0.5$; $T=20000$ unforced, $T=600$ forced at $\omega=2.5$, $\delta \in \{0, 4, 30\}$	4	F01
S1	Blanket additive sweep of the reduced system	ω : 141 values 0.5–4.0 step 0.025; $\delta \in \{0, 0.5, 1.5, 4, 12, 30\}$; $r \in \{0, 0.25, 0.5, 0.9, 1.4\}$; $\sigma \in \{0.5, 1, 2\}$; $\varepsilon \in \{0.02, 0.05, 0.1, 0.2\}$; 4 seeds; $T=3000$, burn 1000	203,040 (33,840 null; 1,440 cells)	sweep_ all.csv; F02–F09
S2	Three-dimensional system, figure scan	$\varepsilon=0.2$, $\sigma=1$; $\tau \in \{0.05, 0.2, 0.5, 1, 2, 5\}$; $\delta \in \{0, 4, 30\}$; ω : 71 values 0.5–4.0 step 0.05; $T=6000$, burn 2000	1,278	F10
S3	Three-dimensional system, tabulated scan	$\varepsilon \in \{0.2, 0.5\}$; $\tau \in \{0.5, 1, 2, 5\}$; $\delta \in \{0, 1, 4, 12, 30\}$; ω : 61 values 1.0–4.0 step 0.05; 4 seeds	9,760	scan3d_ full.csv
S4	Detector calibration; additive plus a parametric reference	$\varepsilon=0.05$, $\sigma=1$, $r=0.5$; ω : 71 values 0.5–4.0 step 0.05; additive $\delta \in \{0, 4, 30\}$, parametric $\delta \in \{0, 1, 4\}$; $T=6000$, burn 2000	426	F10b
S5	δ sweep of the cycle-period distribution at fixed ω	$\omega=2.5$; δ : 41 values 0–4 step 0.1; $r \in \{0.5, 0\}$; 12 realisations; $T=8000$, burn 2000	984	psweep_ r05/r00.npz; F19, F19b, F19c
S6	Two-tone superposition null (no dynamics)	$\omega_0=1$, $\omega=2.5$, $B/A=0.76$; 40 random phases; $t \leq 40000$, $\Delta t=0.2$	40	F20
S7	Modes of the N distribution at the QBO-analogue frequency	$\omega=2.3333$; $\delta \in \{0, 1, 2, 4, 8, 12\}$; $r \in \{0.5, 0\}$; 24 realisations; $T=12000$, burn 3000	288	modes.npz; F21
S8	Every mode against ω , phase-winding definition	ω : 15 values 1.90–3.30 step 0.10; $\delta \in \{1, 2, 4\}$ at $r=0.5$ and $\delta \in \{2, 4\}$ at $r=0$; 12 realisations; $T=6000$, burn 1500	900	allpeaks- .npz, allpeaks- _r0.npz; F22
S9	Fine (ω, δ) zero-crossing sweep, both counting rules	ω : 151 values 1.00–4.00 step 0.02; δ : 81 values 0–4 step 0.05; 3 seeds; $r \in \{0.5, 0\}$; $\varepsilon=0.05$, $\sigma=1$; $T=4000$, burn 1000	12,231 cells $\times$ 3 seeds $\times$ 2 values of r	zc_r05.npz, zc_r00.npz; F23–F29
S10	Branch slope against δ (re-analysis of S9, no new simulation)	δ : 81 values 0–4 step 0.05; $r \in \{0.5, 0\}$; both counting rules	—	slope_vs_ delta.npz; F30

9 Index of figures

Fig.	Source	What it shows	Axes
F01	V1	Integrator validation, three panels	h vs density; period vs density; t vs X
F02	S1	Response spectrum maps at four δ	ω vs response frequency, colour power
F03	S1	The same over a 5×5 grid of (δ, r)	as F02
F06	S1	Welch spectra of 36 individual runs	response frequency vs normalised power
F04	S1	Tracking slope and peak travel	ω vs response frequency; slope vs count; travel vs count
F05	S1	Linear response, dependence on r , spectrum of h	ω vs amplitude; r vs max slope; δ vs band power
F07	—	Four sample paths and their energies	t vs X ; t vs h

Fig.	Source	What it shows	Axes
F08	S1	Six distribution panels	see caption
F09	S1	Phase coherence, map, null comparison, trend	ω - δ map; R_2 vs density; δ vs R_n
F10	S2	Three-dimensional system, three rows $\times$ six τ	ω vs peak, R_2 , λ
F10b	S4	Detector calibration, additive and parametric	ω vs response frequency; ω vs R_2
F19	S5	41 period distributions, $r=0.5$	period vs density (log)
F19b	S5	The same at $r=0$	as F19
F19c	S5	The sweep as maps and band fractions	period vs δ ; δ vs fraction
F20	S5, S6	Simulation against the two-tone null	period vs density; δ vs fraction
F21	S7	N at $\omega=2.333$: densities and mode positions	N vs density; δ vs mode position
F22	S8	Every mode against ω , phase winding	ω vs mode position in N
F23	S9	The two counting rules on one trajectory	t vs X
F24	S9	Median N , its change, and cuts	ω - δ maps; ω vs ΔN
F24b	S9	Median N both rules, plus $ \partial N / \partial \omega $	ω - δ maps
F25	S9	Median N vs ω ; plateau coverage	ω vs N ; δ vs fraction
F27	S9	Integer-band occupancy, both rules	ω - δ maps, colour occupancy
F26	S9	Occupancy minus its unforced value	ω - δ maps, diverging colour
F28	S9	Distribution of N against a smooth envelope	N vs density
F29	S9	Every mode against ω , zero crossings	ω vs mode position in N
F30	S10	Branch slope against δ	δ vs $dN/d\omega$

10 Index of measured values

Quantity	Measured value
<i>Simulation S1 — the blanket sweep</i>	
Total runs / null runs / parameter cells / ω values	203,040 / 33,840 / 1,440 / 141
Tracking slope, 2:1 window: mean, maximum, cells	+0.0023, +0.2017, 1,200
Tracking slope, 3:1 window: mean, maximum	−0.0005, +0.1084
Cells reaching half the 2:1 or 3:1 slope	0 and 0
Total travel of the peak: median, maximum	0.0691, 0.3236
Maximum tracking slope by r (0/0.25/0.5/0.9/1.4)	4.9×10^{-16} / 0.1096 / 0.0979 / 0.2017 / 0.0520
Power of h at 2ω at $\delta = 0$ and $\delta = 30$	0.25% and 31.7%
Forced runs exceeding the ω -matched null in R_2 , R_3	0.47%, 0.74%
Median R_2 : forced, null	0.0092, 0.0142
Largest R_2 anywhere in the null	0.3574
Bimodality coefficient $> 5/9$: forced, forced at $r = 0$, null	44.5%, 31.3%, 1.25%
Runs within 1% of $4\pi/\omega$: forced, null	1.160%, 1.161%
<i>Simulation S3 — three-dimensional system</i>	
Runs	9,760
<i>Simulation S4 — detector calibration</i>	
Tracking slope on $\omega \in [1.8, 2.2]$, additive $\delta = 4, 30$	−0.0314, −0.0204
Tracking slope, parametric reference $\delta = 1, 4$	+0.4979, +0.4979

Quantity	Measured value
Mean R_2 on the same window, additive $\delta = 4, 30$	0.0219, 0.0164
Mean R_2 , parametric reference $\delta = 1, 4$	0.9765, 0.8841
<i>Simulation S7 — $\omega = 2.333$, phase winding, $r = 0.5$</i>	
Weight within ± 0.04 of $N = 1, 2, 3, \omega$ at $\delta = 0$	0.28%, 18.07%, 2.53%, 47.68%
The same at $\delta = 2$	30.27%, 31.65%, 5.06%, 9.58%
The same at $\delta = 4$	53.29%, 21.17%, 2.65%, 3.80%
<i>Simulation S9 — the fine zero-crossing sweep</i>	
Cells / ω values / δ values / seeds	12,231 / 151 / 81 / 3
Cycles per cell: median, minimum	1,480, 1,342
Validation at $\delta = 0$: median N at $\omega = 1.5\text{--}3.5$	1.5017, 2.0023, 2.5029, 3.0034, 3.5040
Typical $ N(r=0.5) - N(r=0) $	0.030
Fraction of ω with $N(\delta) < N(0)$, $r = 0.5$, $\delta = 1\text{--}4$	57.3%, 75.0%, 87.1%, 94.4%
The same at $r = 0$	64.5%, 86.3%, 96.8%, 100%
Mean $N(\delta) - N(0)$, $r = 0.5$, $\delta = 1\text{--}4$	−0.013, −0.082, −0.196, −0.358
The same at $r = 0$	−0.038, −0.196, −0.441, −0.740
Plateau coverage ($ \partial N / \partial \omega < 0.25$), $\delta = 0\text{--}4$	0, 0, 0, 8.1%, 19.4%
Position in ω of the median- $N = 2$ contour, $\delta = 0\text{--}4$	1.998, 2.014, 2.083, 2.162, 2.318
Largest band-1 occupancy at $\delta = 4$ (and at $\delta = 0$, same ω)	0.933 at $\omega = 1.00$ (0.709)
Largest band-2 occupancy at $\delta = 4$ (and at $\delta = 0$)	0.321 at $\omega = 1.96$ (0.415)
Largest band-3 occupancy at $\delta = 4$ (and at $\delta = 0$)	0.364 at $\omega = 3.02$ (0.301)
Slope of the $N \approx 2$ branch, hysteresis, $\delta = 1, 2, 4$	+0.959, +0.523, +0.328
The same, raw	+0.961, +0.540, +0.376
Mean N on the $N \approx 1$ branch, hysteresis, $\delta = 1, 2, 4$	1.090, 1.004, 0.988
Raw vs hysteresis crossings in the window of F23	14 vs 7, 7 rejected; $h_{\text{th}} = 0.55$
<i>Derived quantity S10 — branch slope against δ</i>	
Plateau slope at small δ (four estimators)	0.9555, 0.9538, 0.9568, 0.9559
Onset of departure from the plateau, $r = 0.5$ / $r = 0$	$\delta = 1.50$ / $\delta = 1.20$
Slope at $\delta = 4$ (four estimators)	0.328, 0.376, 0.314, 0.353
Rate of fall over $\delta \in [0.2, 1.1]$	−0.004, −0.010, −0.093, −0.089
Rate of fall over $\delta \in [1.4, 2.0]$	−0.605, −0.620, −0.480, −0.369
Rate of fall over $\delta \in [2.2, 4.0]$	−0.078, −0.074, −0.058, −0.062

In the four-estimator rows the order is: $r = 0.5$ hysteresis, $r = 0.5$ raw, $r = 0$ hysteresis, $r = 0$ raw.